

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)							Semester: II		
Course: Engineering Chemistry							Code: ECBS204		
Teaching Scheme (Hrs/week)				Evaluation Scheme (Marks)					
Lecture	Practical	Tutorial	Credit	CIE	ETE	TW	OR	PR	Total
02	02	-	03	40	60	25	-	-	125
Prerequisites:									
Basic knowledge of volumetric analysis, structure property relationship, classification and properties of polymers, electromagnetic radiation, electrochemical series.									
Course Objectives:									
1. To familiarize the students with the basic phenomenon/concepts of chemistry and its applications in various fields of Engineering. 2. To impart knowledge of technologies involved in water analysis to improve water quality. 3. To learn significance science of corrosion and preventive methods used for minimizing corrosion. 4. To understand structure, properties and applications of speciality polymers and nanomaterials.									
Course Outcomes: After completion of this course, students will be able to -									
CO1	Analyze water softening parameters.								
CO2	Utilize different analytical methods for analysis of various chemical compounds.								
CO3	Understand the mechanism of destruction of metals (corrosion) and effective preventive measures.								
CO4	Explore the knowledge of advanced engineering materials for various engineering applications.								
CO5	Analyze fuel and suggest use of alternative fuels.								
CO6	Familiarize with classification, properties and applications of nanomaterials.								
Course Contents:									
Unit	Description								Duration (Hrs.)
1.	Water Technology: Introduction, Chemical Analysis of Water- Hardness; Temporary and Permanent, Alkalinity (Hydroxide, Carbonate and Bicarbonate), Softening Methods: Zeolite and Demineralization Process, Water Purification: Reverse Osmosis. Simple Numerical on Hardness Determination and Alkalinity Calculation.								5
2.	Instrumental Methods of Analysis: Types of analysis: Quantitative and Qualitative analysis Introduction, Instrumentation and Applications of following methods: Colorimetry, pHmetry (Titration of Strong acid versus Strong base), Conductometry (Titration of Strong acid versus Strong base)								5
3.	Corrosion Science: Introduction, Types of Corrosion-Dry and Wet corrosion, Wet Corrosion Mechanism: Hydrogen Evolution and Oxygen Absorption, Factors affecting rate of corrosion. Methods of prevention of corrosion: Cathodic Protection (Sacrificial Anode), Anodic Protection (Anodizing), Methods to apply Metallic Coatings-Hot dipping, Electroplating.								4

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4.	Engineering Polymers: Polymers: Introduction, Definition of Polymer, Monomer and Functionality of monomers Speciality Polymers: Introduction, Preparation, Properties and Applications of the following polymers: 1. Engineering Thermoplastic: Polycarbonate 2. Conducting Polymer: Polyacetylene Polymer Composites: Introduction, Constituents of composite, Advantages over conventional materials, Applications, Fiber Reinforced Plastic (FRP)-Glass reinforced and Carbon reinforced.	5
5.	Fuels and Combustion: Introduction, Calorific value - Definition, Gross and Net calorific value, Determination of Calorific value: Principle, Construction and Working of Bomb Calorimeter (Simple Numerical), Solid fuel: Coal: Analysis of Coal-Proximate (Simple Numerical). Alternate fuels: Biodiesel and Power alcohol. Hydrogen as future fuel: Production, Advantages, Storage and Applications in Hydrogen fuel cell.	5
6.	Nanomaterials: Introduction, Classification of Nanomaterials Based on Dimensions, Nanoscale materials: Structure, Properties and Applications of Graphene and Quantum dots (semiconductor nanoparticles), Importance of Nanotechnology in engineering applications.	4
TOTAL		28
List of Experiments:		
A. Lab Experiments (Any Seven) - Determination of hardness of water by EDTA method. - Determination of alkalinity of water. - Determination of strength of strong acid using pH meter. - Determination of maximum wavelength of absorption of $\text{CuSO}_4/\text{FeSO}_4/\text{KMnO}_4$, verify Beer's law and find unknown concentration of given sample. - Titration of a mixture of strong acid with strong base using Conductometer. - Preparation of phenol-formaldehyde/urea-formaldehyde resin. - Proximate analysis of coal. - Coating of copper or zinc on iron plate using electroplating. - Determination of the molecular weight of a polymer by using Ostwald's Viscometer.		
B. Demonstration (virtual) (Any One) - Demonstration of effect of environmental conditions on metal by weight loss method. - Synthesis of oxide nanoparticles.		
C. Mandatory visit to chemical industry/research laboratory/water treatment plant.		
Text Books:		
- O.G. Palanna, "Engineering Chemistry", Tata McGraw Hill Education Pvt. Ltd. - Dara S. S., Umare S. A., "Textbook of Engineering Chemistry", 12th Ed, S. Chand & Com Ltd. - Jain and Jain, "Engineering Chemistry", 16th Ed, Dhanpat Rai and Co. (Pvt.) Ltd., Delhi.		

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Reference Books:

1. G. R. Chatwal & S. K. Anand, "Instrumental Methods of Chemical Analysis", Himalaya Publishing House.
2. Dr. Sunita Rattan; A Textbook of Engineering Chemistry; 3rd Ed, S. K. Kataria & Sons, New Delhi
3. V. R. Gowarikar, N. V. Viswanathan, Jayadev Sreedhar, "Polymer Science", Wiley Eastern Limited.
4. Billmeyer F. W., "Textbook of polymer science", John Wiley and Sons.
5. B. Sivasankar, "Engineering Chemistry", Tata McGraw-Hill Education Publishing company Limited.
6. G. L. Hornyak, J. J. Moone, H. F. Tihale, J. Dutta "Fundamentals of Nanotechnology", CRC press.

E-Resources:

MOOC / NPTEL/YouTube Links:

1. NPTEL Course on Corrosion, IISc Bangalore : <http://nptel.ac.in/courses/113108051/>
2. NPTEL Course on Polymer, IIT Kharagpur: <http://nptel.ac.in/courses/104105039/>,
<http://nptel.ac.in/courses/104103071/40>
3. NPTEL Course on Water Technology, IIT Kanpur: <http://nptel.ac.in/courses/105104102/>
4. NPTEL Course on UV-Visible Spectroscopy: <http://nptel.ac.in/courses/102103044/4>
5. NPTEL Course on Energy Sources: <http://nptel.ac.in/courses/103105110/4>
6. NPTEL Course on "Engineering Chemistry-I, <https://nptel.ac.in/courses/122/106/122106028/>
7. NPTEL Course on "Fundamentals of Spectroscopy", NCL, IISER Pune
<https://nptel.ac.in/courses/104/106/104106122/>

Virtual Labs:

1. PICT Pune: <http://chemistryvl.pict.edu/#/>
2. NITK Surathkal: Hardness of water: <https://ee1-nitk.vlabs.ac.in/exp/determination-of-hardness/simulation.html#>
3. NITK Surathkal: Alkalinity of water: <https://ee1-nitk.vlabs.ac.in/exp/determination-of-alkalinity/simulation.html>
4. IIT Hyderabad: Colorimeter, verification of Beer's law, <https://mas-iiith.vlabs.ac.in/exp/beer-law/simulation.html>
5. IIT Kanpur: Preparation of phenol-formaldehyde resin, <http://ebootathon.com/labs/beta/chemistry/EngineeringChemistryLab/exp1/simulation.html>
6. Amrita University: Determination of viscosity average molecular weight polymer, https://pcv-au.vlabs.ac.in/physicalchemistry/Determination_of_ViscosityAverageMolecularWeightofPolymer/